

DONALD WOODWARD

Senior Full-Stack Engineer | Tech Lead | Frontend & Backend Architect | Distributed Systems
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Summary

Senior Software Engineer and Architect with **10+ years of experience** designing, building, and optimizing scalable full-stack architectures across **Fintech, Real Estate, E-commerce, HealthTech, and Education** sectors. Deep expertise encompasses modern application development (**React/Next.js, Angular, React Native**), scalable distributed services (**Node.js, Ruby on Rails, Python**), and high-throughput data engineering utilizing **Kafka and Kubernetes**. A demonstrated leader in generating **significant performance gains** and **bolstering security defenses**, consistently meeting stringent **HIPAA and compliance standards**.

Skills

Language: JavaScript • TypeScript • PHP • Python • Ruby

Frontend: React • Next.js • React Native • Flutter • Angular • Vue.js • Tailwind CSS • Material UI • AntD

Backend: Node.js • NestJS • Express • Laravel • Java • FastAPI • Django • Flask • Ruby on Rails • GraphQL • Rest APIs • WebSocket

Data & Messaging: PostgreSQL • MySQL • MongoDB • Cassandra • Redis • Kafka • RabbitMQ • Snowflake

Cloud & DevOps: AWS (EC2 • ECS • EKS • S3 • RDS • SQS • ElastiCache • CloudFront) • Docker • Kubernetes • CI/CD (GitHub Actions • GitLab)

Testing & Monitoring: Jest • Supertest • SonarQube • Sentry • LogRocket

Security & Auth: OAuth 2.0 • JWT • TLS/mTLS • FHIR

Other: LaunchDarkly • Storybook • Agile/Scrum • Git

Experience

Virtusa

Tech Lead, Senior Full Stack Engineer

01/2025 - 04/2026

Contract

- Lead a cross-functional team of **6+ engineers** developing **AI-powered fintech banking solutions** for enterprise clients, overseeing fullstack development across web **React/TypeScript** and mobile **React Native** platforms.
- Architected intelligent banking features using **OpenAI GPT-4** and **LangChain**, including conversational financial assistants, automated document processing, and natural-language interfaces.
- Built **RAG systems** with vector embeddings for real-time financial document analysis, enabling automated underwriting decisions and compliance reporting while maintaining **PCI DSS** and **SOX** standards.
- Designed and launched cross-platform **React Native mobile banking apps** with biometric authentication, real-time transaction processing, and AI-driven financial insights, serving **100K+ active users** with **99.9% uptime**.
- Led **Python backend development** for core banking microservices, implementing **Node.js** and **GraphQL APIs**, real-time event processing, and machine learning pipelines for **fraud detection** and **credit risk assessment**.
- Established technical leadership practices including architecture reviews, code standards, **CI/CD pipelines**, and mentorship programs, reducing deployment time by **60%** and improving code quality metrics across the team.
- Collaborated with banking clients to translate financial requirements into scalable technical solutions, managing project delivery and ensuring **regulatory compliance**.
- Established a test-first engineering culture using **Jest** and **Supertest** with NestJS testing utilities, achieving **85%+ coverage** across unit and integration tests and catching **90%+** of regressions before production.

Blend

Senior Software Engineer

12/2022 - 12/2024

Full-time

- Led the development of the unified **mortgage and auto loan origination process**, implementing **Native Module Bridging** that cut CPU utilization by **35%** to achieve sustained **60 FPS** fidelity.
- Engineered and deployed the external marketing site and critical consumer-facing pages using **Next.js** and **React**, leveraging **SSR** and **Edge Functions** to boost the target funnel's SEO ranking **from page 3 to page 1** and **slash TTI by 40%**.
- Established and governed a centralized **monorepo component library (150 reusable components)**, which accelerated cross-functional feature delivery by **30%** and reduced UI-related bugs by **18%**.
- Architected a robust **Ruby on Rails** transactional API, successfully processing over **50K+ requests per minute (RPM)** at peak load and achieving an average **P99 latency of 95ms** to support the **React Native** application.
- Revolutionized data access with a **GraphQL API Gateway** to consolidate fetching across **6 distinct microservices**, cutting payload size and frontend over-fetching by **50%** across **10K+ daily queries**.
- Secured the platform by integrating **OAuth 2.0** and **JWTs** for stateless session management, maintaining a documented **99.8% crash-free rate** (via Firebase Crashlytics/Sentry).
- Directed the implementation of **8 primary CI/CD pipelines**, cutting average deployment cycle time from 45 minutes to **15 minutes** and reducing environment setup errors by **50%**.
- Instilled** adherence to security and governance standards by establishing **3 mandatory code review practices** and **auto-generating compliance reports**, ensuring **zero critical audit failures** in two consecutive years.

Experience

Curative Inc

Senior Software Engineer

02/2020 - 11/2022

Full-time

- Collaborated closely with the **CTO and clinical leadership**, leading 6 engineers to rapidly build and scale the nationwide **COVID-19 vaccination platform**, serving **millions** at peak demand.
- Developed **responsive booking interfaces** in **React** and **Next.js** with real-time **REST/GraphQL APIs**, enabling seamless scheduling of first, second, and booster doses across all devices.
- Accelerated patient self-service booking efficiency by **15%** by executing client-side performance optimizations, including memoization and routing logic adjustments, maintaining response time under **200ms**.
- Designed the real-time scheduling backend with **transactional APIs**, **Redis locks**, and **Celery background jobs**, managing **100k+ weekly appointments**.
- Engineered **RabbitMQ pipelines** to reliably synchronize vaccination records across **50+ state registries**, ensuring error-free delivery for auditing and compliance.
- Fortified the API perimeter by implementing and enforcing **TLS** and **Mutual TLS** client authentication and validating all incoming payloads against the industry-standard **FHIR schema**.
- Orchestrated CI/CD pipelines on **AWS** with **Docker** and **GitHub Actions**, delivering **over 120 stable releases per year** with **zero downtime** in high-pressure production environments.
- Enforced mandatory quality gates, sustaining **85% test coverage** through behavior-driven unit **Jest** and integration **Supertest** tests across all critical API routes.
- Collaborated with clinical teams to rapidly design, implement, and deploy adaptive solutions, enabling real-time response to change healthcare requirements.

Valtech

Senior Full Stack Developer

07/2018 - 02/2020

Full-time

- Led the maintenance and feature extension across both modern **React** and legacy **Angular** UIs for core e-commerce features, actively reducing technical debt by **30%** on critical customer paths.
- Implemented **WebSocket APIs** and built corresponding **React** UI components to deliver critical, **sub-100ms** real-time inventory insights directly to customer interfaces.
- Integrated scalable, high-accuracy ML model predictions into the **React** UI layer via **REST** endpoints, powering the Smart Substitutions feature and achieving over **95% accuracy** in product recommendations.
- Accelerated frontend load times by tuning **PostgreSQL** queries and leveraging **Cassandra** for improved distributed read performance, resulting in a **60% speed increase** for critical API calls.
- Engineered an automated **LaunchDarkly-based experimentation system** within the frontend codebase to validate performance and feature rollouts, facilitating the concurrent execution of **100 Online Controlled Experiments (OCEs)**.
- Authored technical documentation, including API specifications for the **WebSocket** inventory system and operational runbooks for the **A/B testing framework**, improving onboarding efficiency by **20%** for new engineers.

Remind

Software Engineer

03/2016 - 06/2018

Full-time

- Optimized **9 Flask and Django microservices** that powered the two-way text messaging platform, reliably facilitating **billions of messages annually** while maintaining **99.99% uptime**.
- Pioneered backend logic for complex asynchronous tasks, Scheduled Messages, and executed the proprietary **Contact Shielding** feature, guaranteeing **100% user privacy** via PII abstraction.
- Developed frontend features on the teacher dashboard, creating directive-based components for **8 major teacher features** that streamlined class management and bulk communication scheduling.
- Constructed real-time **Angular** components for the Stamps Polls feature, leveraging **WebSockets** and **PostgreSQL pub-sub** functionality to deliver immediate, interactive classroom feedback, supporting **1M concurrent users**.
- Designed and exposed APIs for the Multi-Media Sharing feature, ensuring seamless integration with native **iOS and Android** clients for file and link exchange.

RealtyMogul

Full Stack Developer

01/2015 - 03/2016

Full-time

- Revamped the Investor and Borrower Dashboards, launching **over 10 high-impact features** that boosted investor engagement by **25%**.
- Optimized the core **Ruby on Rails** backend by refactoring **over 30 critical ActiveRecord queries** and implementing multi-level caching strategies, achieving a **90% cache hit rate** on key API routes.
- Executed critical schema and application logic updates in **Ruby on Rails** to integrate **Regulation A+** investor types, completing the project within a **3-month window** to ensure platform compliance.
- Instituted an extensive unit and integration test suite for complex financial calculations, achieving **95% coverage** on all financial logic, ensuring accurate **K-1 tax documentation** annually.

Education

National University

Bachelor of Science in Computer Science

01/2010 - 06/2014